

Executive Summary & Technical Report: HR Employee Attrition Analysis

Project Overview

This report documents the end-to-end process of building the **HR Employee Attrition Analysis Dashboard** using Power BI. The goal of this analysis is to evaluate workforce turnover dynamics, identify high-risk employee segments, assess the impact of overtime and job satisfaction, and provide actionable retention strategies across a total population of **1,470 employees**.

1. Data Cleaning & Transformation (Power Query)

Before visual layout and dashboard construction, raw HR records underwent systematic preprocessing in Power Query Editor:

- **Null & Duplicate Auditing:** Validated employee records for completeness across demographic attributes (Department, Gender, Marital Status, Business Travel) and performance metrics.
- **Data Type Enforcement:** Explicitly assigned proper data types:
 - **Employee Counts & Income:** Whole Number / Decimal (Monthly Income, Years at Company, Age)
 - **Categorical Fields:** Text (Department, Job Role, Overtime, Business Travel)
- **Demographic Binning:** Created age demographic groups (Under 25, 25–34, 35–44, 45–54, 55+) to analyze turnover risk across generational cohorts.

2. Data Modeling & DAX Measures

Custom DAX (Data Analysis Expressions) measures were created to drive key metrics and interactive visuals:

- **Total Headcount**

\$\$\text{Total Employees} = \text{COUNTROWS('HR_Data')}\$\$

- **Total Departures (Attrition)**

\$\$\text{Employees Left} = \text{CALCULATE(COUNTROWS('HR_Data'), 'HR_Data'[Attrition] = "Yes")}\$\$

- **Attrition Rate Percentage**

$$\text{Attrition Rate} = \frac{\text{Employees Left}}{\text{Total Employees}} \times 100$$

- **Average Monthly Income**

$$\text{Avg Monthly Income} = \frac{\sum (\text{Monthly Income})}{\text{Total Employees}}$$

3. Key Analytical Insights

- **Turnover Baseline:** Out of 1,470 total employees, **237 employees left the company**, resulting in an overall **attrition rate of 16.1%** with an average monthly salary of **\$6,503**.
- **Departmental Concentration:** The **Research & Development** department accounts for the highest volume of departures (**133**), followed by **Sales (92)** and **Human Resources (12)**.
- **Role Vulnerability:** Top roles experiencing turnover include **Laboratory Technicians (62)**, **Sales Executives (57)**, and **Research Scientists (47)**.
- **Overtime Driver:** Employees working **overtime represent 46% (110)** of all departures, identifying heavy workload as a primary attrition risk factor.
- **Tenure Risk Window:** Attrition peaks heavily during the **first 1 to 3 years at the company** before steadily declining for longer-tenured employees.

4. Strategic Recommendations

- **Early-Tenure Engagement:** Introduce structured 30-60-90 day onboarding check-ins and mentorship to mitigate turnover during the high-risk 1–3 year tenure window.
- **Overtime Rebalancing:** Audit workload distribution in R&D and Sales to control overtime demands and prevent employee burnout.
- **Targeted Career Pathways:** Develop clear career advancement tracks for staff aged **25–34** in technical and sales roles to boost long-term retention.